

Definition and first terms

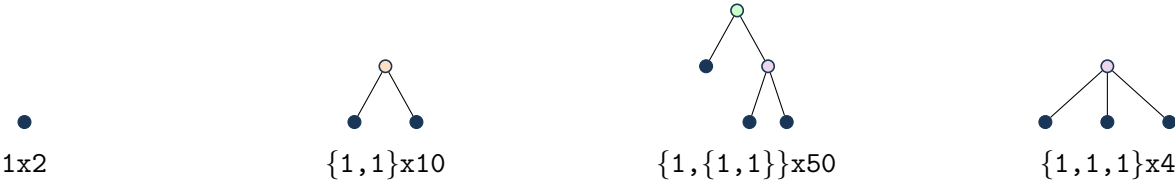
$(5 - 4 x) A(x) = 4 + A(x)^3$, with the origin-normalized solution.

Normalization/grammar: $A(x)=1+(2)*T(x)$; $T=x+2*x*T+3*T^2+2*T^3$.

$a(0), \dots, a(7) = 1, 2, 10, 84, 882, 10380, 130916, 1729960$.

Kernel used throughout: $\rho(u) = u \frac{1 - 3 u^1 - 2 u^2}{1 + 2 u}$, so $x = \rho(T(x))$.

Typogeometry



colored arity geometry; node color distinguishes constructors. Filled endpoints are true leaves; open endpoints are false/empty leaves. For colored grammars, color is genuine constructor data and is not silently converted into a positional zero pattern.

Typogeometric constructor codes and multiplicities: $\Delta_2: 3$, $\Delta_2^{\text{marked}}: 2$, $\Delta_3: 2$.

Coefficient and contour extraction

$$\mathcal{K}_{n,h} = \{(k_1, \dots, k_2) \in \mathbb{Z}_{\geq 0}^2 : h + \sum_{j=1}^2 j k_j = n - 1\}, \quad K = \sum_{j=1}^2 k_j,$$
$$a(n) = \frac{2}{n} \sum_{h=0}^{n-1} \binom{n}{h} \sum_{\mathbf{k} \in \mathcal{K}_{n,h}} \binom{n+K-1}{K} \binom{K}{k_1, \dots, k_2} 3^{k_1} 1^{k_2}.$$
$$a(n) = \frac{2}{2\pi i n} \oint_{\gamma} \frac{du}{\rho(u)^n}, \quad n \geq 1.$$

Recurrence

$\sum_{r=0}^4 P_r(n) a(n+r) = 0$, $n \geq 1$, with

$$\begin{aligned} P_0(n) &= 256 n^3 + 128 n^2 - 128 n \\ P_1(n) &= -1280 n^3 - 3360 n^2 - 2720 n - 640 \\ P_2(n) &= 2400 n^3 + 11400 n^2 + 17800 n + 9200 \\ P_3(n) &= -1568 n^3 - 11590 n^2 - 28158 n - 22500 \\ P_4(n) &= 85 n^3 + 765 n^2 + 2210 n + 2040 \end{aligned}$$

Differential equation

The scalar linear ODE has order 3. It is obtained from the recurrence by the standard Euler-operator substitution. Exact coefficients and boundary polynomial: `runs/A244627-direct-x-pilot/case.json#/ode_for_A`.

Matrices, telescoper certificate, and checks

No compact matrix tuple is shown; see the embedded exact reduction data.

Certificate.

$R(n, u, x) = N/D$ (exact 275-character numerator in embedded payload).

Telescoping identity: $\sum_r P_r(n) H_{n+r}(u) = \partial_u (R(n, u) H_n(u))$.

Matrix dimensions: 6×6 ; remainder 2×3 , rank 2, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.